Beyond Ingress

Mastering the Kubernetes Gateway API

Wolfgang Ofner
Senior Cloud Architect

Agenda



Expose Pods

Gateway API

Routing

Gateway API vs. Ingress

Wolfgang Ofner

Freelance Cloud Architect, Toronto, Canada

Focus on Azure, Kubernetes, and DevOps

programmingwithwolfgang.com

wolfgangofner

YouTube @programmingwithwolfgang



Microsoft MVP Most Valuable Professional

Microsoft CERTIFIED CYBERSECURITY ARCHITECT EXPERT

Microsoft CERTIFIED AZURE SOLUTIONS ARCHITECT EXPERT

Microsoft CERTIFIED DEVOPS ENGINEER EXPERT

Microsoft CERTIFIED TRAINER MCT

Expose Pods

Expose Pods

Service with type Load Balancer

Ingress

Gateway API

Ingress and Gateway API Features

- One entry point
- Better routing
- Automate TLS certificates with Cert-Manager
- TLS termination

Ingress

Ingress Controller:

- Nginx
- Traefik
- Application Gateway Ingress Controller (AGIC)

Ingress Downsides

Ingress controllers have a lot of control how traffic is routed

Annotations are unique to the used Ingress controller (e.g. Nginx) → vendor lock-in

No modularity

No real Separation of Concerns

Gateway API

Gateway API

Components are modular and split into multiple parts

Gateway Class configures proxy (similar to Ingress Class)

Gateway Class depends on Gateway Controller

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```



```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: GatewayClass
metadata:
  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
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apiVersion: gateway.networking.k8s.io/v1
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  name: envoy
spec:
  controllerName: gateway.envoyproxy.io/gatewayclass-controller
```

Gateway

Gateway contains listeners

Links to TLS certificates

Namespaced

Gateway consumes global Gateway Class

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
  name: Gateway
  namespace: gateway-ns
  annotations:
    cert-manager.io/cluster-issuer: clusterissuer
spec:
  gatewayClassName: envoy
  listeners:
  - name: http-listener
    port: 80
    protocol: HTTP
    allowedRoutes:
      namespaces:
        from: All
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: Gateway
metadata:
  name: Gateway
  namespace: gateway-ns
  annotations:
    cert-manager.io/cluster-issuer: clusterissuer
spec:
  gatewayClassName: envoy
  listeners:
  - name: http-listener
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    protocol: HTTP
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    cert-manager.io/cluster-issuer: clusterissuer
spec:
  gatewayClassName: envoy
  listeners:
  - name: http-listener
    port: 80
    protocol: HTTP
    allowedRoutes:
      namespaces:
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  gatewayClassName: envoy
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  name: Gateway
  namespace: gateway-ns
  annotations:
    cert-manager.io/cluster-issuer: clusterissuer
spec:
  gatewayClassName: envoy
  listeners:
  - name: http-listener
    port: 80
    protocol: HTTP
    allowedRoutes:
      namespaces:
        from: All
```

```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```

```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```



```
- name: traefik-https-listener
  port: 443
  protocol: HTTPS
  hostname: demo.programmingwithwolfgang.com
  allowedRoutes:
    namespaces:
      from: All
  tls:
    certificateRefs:
      - group: ""
        kind: Secret
        name: demo-secret
```

ListenerSet

Allow teams to configure their listeners

Automate TLS certificate management

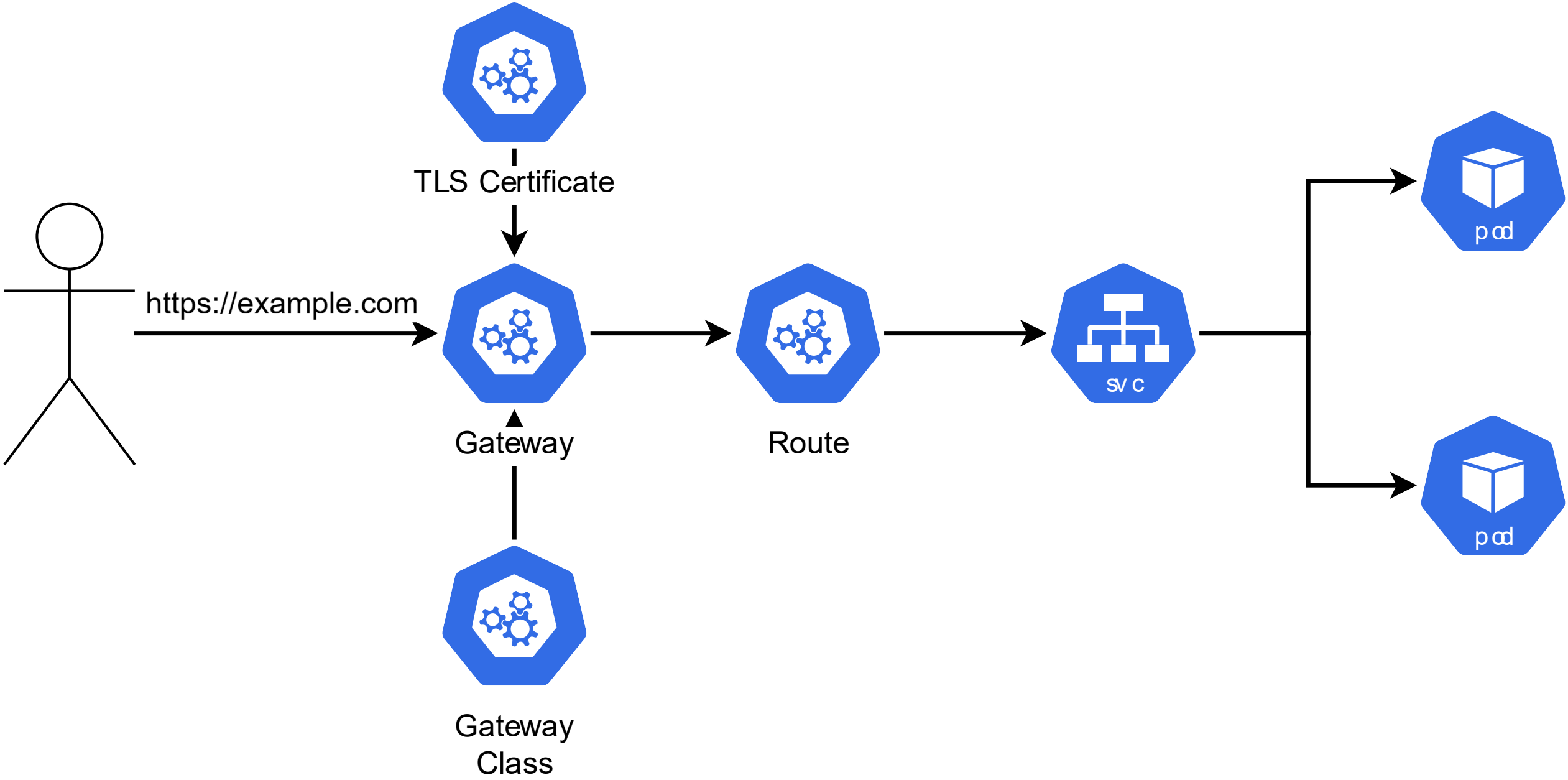
Perfect for multitenancy

Reduces the blast radius

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: ListenerSet
metadata:
  name: demo-listenerset
  namespace: demo-app-ns
spec:
  parentRef:
    group: gateway.networking.k8s.io
    kind: Gateway
    name: Gateway
    namespace: gateway-ns
  listeners:
    - name: http-listener
      protocol: HTTP
      port: 80
      hostname: myapp.programmingwithwolfgang.com
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: ListenerSet
metadata:
  name: demo-listenerset
  namespace: demo-app-ns
spec:
  parentRef:
    group: gateway.networking.k8s.io
    kind: Gateway
    name: Gateway
    namespace: gateway-ns
  listeners:
    - name: http-listener
      protocol: HTTP
      port: 80
      hostname: myapp.programmingwithwolfgang.com
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: ListenerSet
metadata:
  name: demo-listenerset
  namespace: demo-app-ns
spec:
  parentRef:
    group: gateway.networking.k8s.io
    kind: Gateway
    name: Gateway
    namespace: gateway-ns
  listeners:
    - name: http-listener
      protocol: HTTP
      port: 80
      hostname: myapp.programmingwithwolfgang.com
```


TLS Certificate
https://example.com
Gateway
Route
svc
pod
pod
Gateway
Class

Routing

Routing

Listeners can listen on different ports

Different protocols:

- HTTP
- HTTPS
- TLS
- TCP
- UDP

Routes

Uses standards instead of annotations

Routes:

- HTTPS and HTTP
- TCP and UDP
- GRPC
- TLS

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```



```yaml
apiVersion: gateway.networking.k8s.io/v1
kind: HTTPRoute
metadata:
  name: traefik-http-route
  namespace: traefik-ns
spec:
  parentRefs:
    - name: gateway
      namespace: gateway-ns
  hostnames:
    - "demo.programmingwithwolfgang.com"
  rules:
    - matches:
      backendRefs:
        - name: traefik
          port: 80
```

Routing

Routing options:

- Path
- Header, e.g. User-Agent
- Query
- HTTP method
- Redirect request

```
rules:
  - matches:
      - path:
          type: PathPrefix
          value: /my/path
    filters:
      - type: URLRewrite
        urlRewrite:
          path:
            type: ReplaceFullPath
            replaceFullPath: /replaced
    backendRefs:
      - name: app-one
        port: 80
  - backendRefs:
    - name: app-two
      port: 80
```

```
rules:
  - matches:
      - path:
          type: PathPrefix
          value: /my/path
    filters:
      - type: URLRewrite
        urlRewrite:
          path:
            type: ReplaceFullPath
            replaceFullPath: /replaced
    backendRefs:
      - name: app-one
        port: 80
  - backendRefs:
    - name: app-two
      port: 80
```

```
rules:
  - matches:
      - path:
          type: PathPrefix
          value: /my/path
    filters:
      - type: URLRewrite
        urlRewrite:
          path:
            type: ReplaceFullPath
            replaceFullPath: /replaced
    backendRefs:
      - name: app-one
        port: 80
  - backendRefs:
    - name: app-two
      port: 80
```

```
rules:
  - matches:
      - path:
          type: PathPrefix
          value: /my/path
    filters:
      - type: URLRewrite
        urlRewrite:
          path:
            type: ReplaceFullPath
            replaceFullPath: /replaced
    backendRefs:
      - name: app-one
        port: 80
  - backendRefs:
    - name: app-two
      port: 80
```

Gateway API vs Ingress

Gateway API vs Ingress

Gateway API might not replace Ingress

Ingress is very easy to manage

- Especially in smaller teams

Gateway might be better for multi-team setups due to modularity

Beyond Ingress

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